Multiple testing considerations

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Warmup: xkcd comic

- Read the provided xkcd comic
- Do you think that green jelly beans are truly related to acne? If not, why did the scientists see a significant p-value for green jelly beans, but not for other colors?

Motivation: differential gene expression

Suppose a biologist is interested in identifying genes which are *differentially expressed* under different biological treatments. The biologist observes 10 subjects under treatment A, and 10 subjects under treatment B. Gene expression measurements $X_{i,j}$ (treatment A) and $Y_{i,j}$ (treatment B) are recorded for 1000 different genes ($i = 1, ..., 1000$, $j = 1, ..., 10$).

For each gene i, the biologist tests $H_{0i} : \mu_{i,A} = \mu_{i,B}$, rejecting when the p-value is below a threshold α.

If H_0 is actually true for all 1000 genes, how many false positives do we expect?

$\alpha \cdot 1000$

Motivation

In what other settings might we test many hypotheses?

Outcomes for testing multiple hypotheses

Suppose we test m hypotheses $H_{0,1}, \ldots, H_{0,m}$, of which m_0 are truly null (i.e. $H_{0,i}$ is true).

Possible outcomes

	H_0 true	H_0 false	
Reject	V		
Fail to reject			
	m_0	$m - m_0$	m

Want to control V, the # of type I errors (false positives)

<u>Familywise error rate</u>:

$$FWER = P(V \geq 1)$$

(probability of at least one type I error)

FWER

Definition: Suppose we test m null hypotheses $H_{0,1}, ..., H_{0,m}$. The *family-wise error rate* is the probability of making *at least one* type I error:

$$FWER = P\left(\bigcup_{i:H_{0,i} \text{ is true}} \{\text{reject } H_{0,i}\}\right)$$

Suppose all m tests are independent. For each test, we reject if the corresponding p-value $p_i < \alpha$.

$$FWER = 1 - P\left(\bigcap_{i:H_{0,i} \text{ true}} \{\text{fail to reject } H_{0,i}\}\right)$$

$$= 1 - (1-\alpha)^{m_0} \quad \text{(independence)}$$

As $m_0 \uparrow$, $FWER \to 1$

In practice: don't know m_0, use m to get upper bound:

$$FWER \leq 1-(1-\alpha)^m$$

The Sidak correction

$$FWER = P\left(\bigcup_{i:H_{0,i} \text{ is true}} \{\text{reject } H_{0,i}\}\right)$$

If all m hypotheses are independent, at what threshold α^* should we reject each test, so that FWER $\leq \alpha$?

independence: $\text{FWER} \leq 1-(1-\alpha^*)^m$ (exact equality when $m_0 = m$)

$\Rightarrow \quad 1-(1-\alpha^*)^m = \alpha$

$\Rightarrow \quad \alpha^* = 1-(1-\alpha)^{1/m}$

Example: $m = 100$, $\alpha = 0.05 \Rightarrow \alpha^* \approx 0.00051$

<u>Sidak correction</u>: Reject $H_{0,i}$ if $p_i < 1-(1-\alpha)^{1/m}$

As $m \uparrow$, $1-(1-\alpha)^{1/m} \to 0$, so we have to be <u>really</u> stringent as we test more hypotheses

The Bonferroni correction

Sidak correction uses independence. Can we get a bound that always works?

$$FWER = P\left(\bigcup_{i:H_{0,i} \text{ is true}} \{\text{reject } H_{0,i}\}\right)$$

Suppose we reject each $H_{0,i}$ if $p_i < \alpha^*$

$$P\left(\bigcup_{i:H_{0,i} \text{ true}} \{\text{reject } H_{0,i}\}\right) \le \sum_{i:H_{0,i} \text{ true}} P(\text{reject } H_{0,i}) \le \sum_{i:H_{0,i} \text{ true}} \alpha^* = m_0 \alpha^*$$

If we knew m_0, set $\alpha^* = \frac{\alpha}{m_0}$. But we don't know m_0

<u>Bonferroni correction</u>: reject $H_{0,i}$ if $p_i < \frac{\alpha}{m}$

$$\Rightarrow FWER \le \frac{m_0}{m} \cdot \alpha$$

(Bonferroni correction is conservative)

$$\frac{\alpha}{m} < 1 - (1-\alpha)^{1/m}$$

so Bonferroni method is more conservative than Sidak (but doesn't use independence)

Holm's procedure

Suppose we test 5 hypotheses, and observe p-values 0.4, 0.01, 0, 0, 0. Does it still seem reasonable to use the Bonferroni cutoff $\alpha/5$ for each test?

"Ideal" Bonferroni correction: $\frac{\alpha}{m_0}$

Since we don't know m_0, Bonferroni use $\frac{\alpha}{m} \leq \frac{\alpha}{m_0}$

more stringent

$\Rightarrow$ less power if H_A true

<u>Idea</u>: order p-values $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$

First test: if $p_{(1)} < \frac{\alpha}{m}$ (Bonferroni threshold)

then reject $H_{0(1)}$, and continue to $p_{(2)}$

Now check if $p_{(2)} < \frac{\alpha}{m-1}$ ($m-1$ tests remaining)

Continue: reject $H_{0(i)}$ if $p_{(i)} < \frac{\alpha}{m-i+1}$

As soon as $p_{(i)} > \frac{\alpha}{m-i+1}$, <u>stop</u> (fail to reject all remaining H_0)

Holm's procedure

Suppose we test m null hypotheses $H_{0,1}, ..., H_{0,m}$. Let p_i be the corresponding p-value for test i.

- Order the p-values $p_{(1)} \leq p_{(2)} \leq \cdots \leq p_{(m)}$
- Let $i^* = \min \left\{ i : p_{(i)} > \frac{\alpha}{m-i+1} \right\}$
- Reject $H_{0,(i)}$ for all $i < i^*$

Claim: Holm's procedure controls FWER at level α